Bill shape distinguishes Antarctic penguin species

The Day-1 result — one source, a branded Typst PDF

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1 Introduction

The base-R `penguins` dataset records body measurements for 342 penguins of three species collected at Palmer Station, Antarctica (Gorman et al., 2014). It reached the R community first through the **palmerpenguins** package (Horst et al., 2020). Here we ask a simple question: **does bill shape tell the species apart?** We render the answer as a branded PDF with **Typst**, a modern typesetting system bundled with Quarto that does not require LaTeX.

2 Species at a glance

Table 1

Penguins of the Palmer Archipelago

Mean measurements per species

		Bill (mm)		
Species	**N**	**Length**	**Depth**	**Body mass (g)**
Adelie	151	38.8	18.3	3 701
Gentoo	123	47.5	15.0	5 076
Chinstrap	68	48.8	18.4	3 733

Of the three species, Gentoo has the highest mean body mass (5 076 g).

3 Bill length versus depth

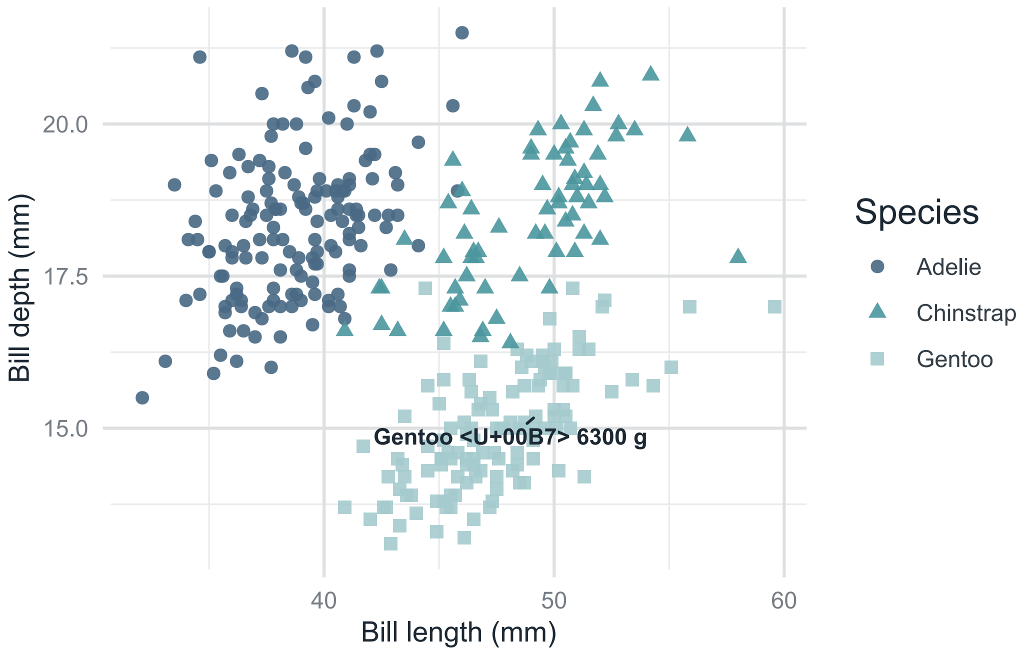

Figure 1: Bill length versus depth by species. The single heaviest penguin is labeled.

The three species separate cleanly in bill shape. This is the same figure from the HTML report, now shown in the branded PDF.

4 Conclusion

One `.qmd`, one `_brand.yml`, and `format: typst` give a typeset document with citations. They apply the RaukR typography and palette to the document, tables, and figures.

References

Gorman, K. B., Williams, T. D., & Fraser, W. R. (2014). Ecological sexual dimorphism and environmental variability within a community of Antarctic penguins (genus Pygoscelis). PLoS ONE, 9(3), e90081. https://doi.org/10.1371/journal.pone.0090081

Horst, A. M., Hill, A. P., & Gorman, K. B. (2020). palmerpenguins: Palmer Archipelago (Antarctica) penguin data. https://doi.org/10.5281/zenodo.3960218